Initial Project Planning Template

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The Smart Lender project was planned using an Agile approach, where the development process was divided into multiple sprints to ensure systematic implementation and continuous progress. The project began with environment setup and dataset analysis, followed by data preprocessing, feature engineering, machine learning model development, and web application integration. Each sprint focused on a specific functional requirement, allowing individual components to be designed, implemented, tested, and refined before moving to the next phase.
The project plan also included estimating story points, assigning priorities, and scheduling sprint durations to effectively manage development time and resources. This structured planning approach ensured that both the machine learning model and the Flask-based web application were developed incrementally, resulting in a reliable, scalable, and user-friendly loan approval prediction system that meets the project objectives

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	**1. Environment Setup & Package Installation**	USN-1	As a developer, I can configure the Python environment and install essential libraries (NumPy, Pandas, Scikit-learn, Flask) to support the ML workflow.	1	High	Sai Geethika Bourothu	25-06-2026	28-06-2026
Sprint-1	**2. Dataset Collection & Understanding**	USN-2	As a data scientist, I can download the dataset and analyze features like Gender and Applicant Income to understand factors influencing loan eligibility.	2	High	Mohana Swaroop Kukkala	25-06-2026	28-06-2026
Sprint-1	**3. Data Visualization & Analysis**	USN-3	As a data scientist, I can perform exploratory data analysis (EDA) using Matplotlib and Seaborn to generate count plots and distribution charts.	2	Medium	Mohana Swaroop Kukkala	25-06-2026	28-06-2026
Sprint-1	**4. Data Preprocessing & Feature**	USN-4	As a data scientist, I can prepare the dataset by filling numerical missing values using	3	High	Mohana Swaroop	25-06-2026	28-06-2026

	Engineering		mean/mode and encoding categorical variables into numerical format.			Kukkala		
Sprint-2	**5. Machine Learning Model Building**	USN-5	As a data scientist, I can develop and train multiple classification models (Decision Tree, Random Forest, KNN, XGBoost) and select the best-performing one.	3	High	Suvvari Ankitha	29-06-2026	03-07-2026
Sprint-2	**6. Building the Flask Web Application**	USN-6	As a backend developer, I can develop an interactive Flask web application that contains a Home Page Introduction and User Input Forms.	2	High	Sai Geethika Bourothu	29-06-2026	03-07-2026
Sprint-2	**6. Building the Flask Web Application**	USN-7	As a user, I can submit my data through the loan eligibility prediction interface and instantly view my Prediction Result Display.	2	High	Suvvari Ankitha	29-06-2026	03-07-2026